BOOPALAN PICHANDI

Full-Stack FinTech Architect | AI Agent Developer | Secure Payments & Cloud Platforms

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PROFESSIONAL SUMMARY

Full-Stack FinTech Architect and AI Agent Developer with 5+ years of progressive experience designing, securing and scaling regulated financial platforms. Technical owner of RhoFi, a SEBI-registered investment research platform (INH000025090), across system architecture, security implementation and end-to-end delivery: React.js UI, Python/Node.js microservices, secure multi-gateway payments (Razorpay, PhonePe, PayU), Kafka/Pub/Sub data pipelines, Kubernetes on AWS/GCP, and LLM-powered AI agents; currently building an MCP-based algorithmic trading agent. Progressed from Full Stack Developer to DevSecOps Engineer to Platform Architect, with a strong track record of turning complex financial requirements into secure, high-performance systems that deliver measurable outcomes.

TECHNICAL SKILLS

Architecture & System Design	Microservices, Event-Driven Architecture, API Gateway & API Design, High Availability, Performance Tuning
Security	OAuth2/JWT, RBAC, Encryption in Transit & at Rest, Secrets Management, WAF & Rate Limiting, Audit Logging, SAST / Dependency / Container Scanning
Payments	Razorpay, PhonePe, PayU, UPI, Subscription Billing, Webhook Verification, Idempotency, Reconciliation, PCI-DSS Scope Reduction
AI / LLM Systems	AI Agents, Model Context Protocol (MCP) Servers, Tool Calling, RAG, Guardrails, LLM Integrations, Algo-Trading Automation
Languages & Frameworks	JavaScript (ES2020+), TypeScript, Python, Node.js, React.js, React Native, Flask, FastAPI, REST, GraphQL
Cloud & Containers	AWS (EC2, S3, RDS, Lambda), Google Cloud Platform, Docker, Kubernetes, Terraform, Zero-Downtime Deployments
DevSecOps & CI/CD	Git, GitHub Actions, GitLab CI, SonarQube, Automated Pipelines, SDLC Compliance
Data & Streaming	PostgreSQL, MongoDB, Redis, Apache Kafka, Google Cloud Pub/Sub, Apache Beam, Query Optimization
Observability & Notifications	ELK Stack, Grafana, Prometheus, Centralized Logging; WhatsApp, Push, Email & In-App Alerts

PROFESSIONAL EXPERIENCE

RhoFi Financial Technologies — Bengaluru **Mar 2024 – Present**

Full-Stack FinTech Architect & AI Agent Developer

- Architected and lead end-to-end development of **RhoFi**, a SEBI-registered research platform for stocks, F&O, mutual funds, IPOs and bonds, with backtested strategies and real-time market insights; primary technical decision-maker across all engineering verticals.
- Designed the platform architecture: event-driven microservices on **AWS and GCP**, **Kafka / Google Cloud Pub/Sub** pipelines for live market data and transaction events, and **Docker + Kubernetes** with horizontal scaling and zero-downtime releases.
- Implemented the security model across every layer: **OAuth2/JWT** with role-based access, encryption in transit and at rest, vault-managed secrets, WAF and rate limiting, and centralized audit logging supporting SEBI obligations.
- Integrated **Razorpay, PhonePe and PayU** behind a single payment service for subscription billing, with server-side order creation, hosted checkout to minimise PCI-DSS scope, and gateway failover.
- Secured the payment flow end-to-end: Razorpay **HMAC-SHA256** signature checks, PhonePe **SHA-256 checksums**, PayU **SHA-512 reverse-hash** verification, amount tamper checks, verified webhooks processed **idempotently via Kafka**, and status-API reconciliation before activation.
- Built **AI agents** and an LLM-powered financial assistant that lets users query portfolio data, market mood and strategies, using scoped read-only tools, retrieval and compliance guardrails.
- **Currently building** an **MCP (Model Context Protocol) server for algorithmic trading**: typed agent tools for trade signals, entry validation, risk checks and Zerodha Kite order placement, with server-side risk limits, a kill switch and a paper-trading mode.

- Engineered React.js frontends and Python (Flask/FastAPI) and Node.js services for real-time financial data; integrated the **Zerodha/Kite** broker API.
- Built a WhatsApp, push and email notification system for trade alerts, portfolio updates and market events on the event-driven Pub/Sub pipeline.
- Built CI/CD and DevSecOps automation with GitHub Actions / GitLab CI and integrated security scanning, **reducing deployment cycles by 60%**.
- Deployed the ELK Stack for centralized observability; designed PostgreSQL and NoSQL architectures for large financial datasets with query tuning and AWS RDS connection pooling.

HawkMonk FinTech — Coimbatore **Nov 2022 – Feb 2024**

DevSecOps Engineer

- Designed and maintained enterprise-grade CI/CD pipelines for fintech applications across staging and production, **improving release velocity by 50%**.
- Enforced DevSecOps practices across the full SDLC, integrating security scanning, vulnerability management and compliance controls into every release pipeline.
- Managed containerized deployments with Docker, improving reliability and **achieving 99.9% uptime SLA** targets.
- Implemented centralized monitoring and logging, **reducing MTTR for production incidents by 40%**.

HawkMonk FinTech — Coimbatore **May 2021 – Nov 2022**

Full Stack Developer

- Developed responsive, production-grade web applications with React.js and RESTful APIs, from UI design through backend integration and deployment.
- Built scalable backend services in Python (Flask) and Node.js, integrating third-party APIs and financial data sources with robust error handling.
- Architected and optimized PostgreSQL schemas and queries, **improving response times by 35%** for growing user and transaction loads.

KEY PROJECT HIGHLIGHTS

- **RhoFi Investment Research Platform**: SEBI-registered platform integrating technical and fundamental research, mutual fund, IPO and bond analytics, backtesting, real-time portfolio tracking, payments and Zerodha/Kite connectivity. Sole technical owner from architecture to production.
- **Secure Multi-Gateway Payments**: One payment service in front of Razorpay, PhonePe and PayU, with server-side orders, signature and hash verification, idempotent webhook processing through Kafka, reconciliation and gateway failover.
- **AI Agents & LLM Insight Engine**: Financial assistant and agents for portfolio and market Q&A with tool calling, RAG and guardrails, plus LLM automation that speeds up analyst research workflows.
- **MCP Algo-Trading Agent (in progress)**: MCP server exposing get_signals, validate_entry, check_risk and place_order tools to an AI agent; every order passes a code-enforced risk gate before it reaches the broker API.
- **Real-Time Notification Platform**: Multi-channel WhatsApp, push, email and in-app alerts for market events and portfolio triggers, powered by Kafka, Google Pub/Sub and event-driven microservices.

EDUCATION

Master of Computer Applications (MCA) **2017 – 2020**

College of Engineering, Guindy — Anna University, Chennai

B.Sc. Mathematics **2014 – 2017**

Muthurangam Government Arts College — Thiruvalluvar University, Vellore